4. Particular Integral (y_p): Trigonometric Case

When the right-hand side of a linear differential equation contains sine or cosine functions ($F(x) = \sin ax$ or $\cos ax$), we use a specific substitution rule involving the square of the differential operator (D^2).

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1. The Method ($D^2 \to -a^2$)

Given a differential equation $f(D^2)y = \sin ax$ (or $\cos ax$):

The Particular Integral is:

$$y_p = \frac{1}{f(D^2)} \sin ax$$

1.1 The Rule

Replace every instance of D^2 with $-a^2$.

$$y_p = \frac{1}{f(-a^2)} \sin ax$$

> **Important Warning:** The negative sign is **outside** the square. * If $a = 2$, then D^2 becomes $-(2^2) = -4$. * It does **NOT** become $(-2)^2 = +4$.

1.2 Handling Remaining D Terms (Rationalization)

If replacing D^2 leaves a D term in the denominator (e.g., $D - 3$), you cannot substitute $-a^2$ for D directly. Instead: 1. Multiply the numerator and denominator by the conjugate (e.g., $D + 3$). 2. This creates a difference of squares ($D^2 - 9$) in the denominator. 3. Replace the new D^2 with $-a^2$ again. 4. Apply the numerator operator to the function.

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2. The Case of Failure ($f(-a^2) = 0$)

If replacing D^2 with $-a^2$ causes the denominator to become zero, the method fails. We apply the standard failure rule:

1. Multiply the numerator by x.
2. Differentiate the denominator with respect to D ($f'(D)$).
3. Try the substitution again.

$$y_p = x \cdot \frac{1}{f'(D)} \sin ax$$

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3. Solved Questions

Here are three distinct cases from the lecture notes: a standard substitution, a rationalization case, and a case of failure.

Question 1: Standard Substitution

Solve for y_p: $(D^2 + 4)y = \sin 3x$

Solution:

$$y_p = \frac{1}{D^2 + 4} \sin 3x$$

Step 1: Identify a Here, $a = 3$. Therefore, $-a^2 = -(3^2) = -9$.

Step 2: Substitute $D^2 = -9$

$$y_p = \frac{1}{-9+4} \sin 3x$$

$$y_p = \frac{1}{-5} \sin 3x$$

Final Answer:

$$y_p = -\frac{1}{5} \sin 3x$$

Question 2: Rationalization Required

Solve for y_p**:** $(D^2 + D + 1)y = \cos 2x$

Solution:

$$y_p = \frac{1}{D^2 + D + 1} \cos 2x$$

Step 1: Identify a **and Substitute** Here, $a = 2$. So, $D^2 \to -4$.

$$y_p = \frac{1}{-4 + D + 1} \cos 2x$$

$$y_p = \frac{1}{D - 3} \cos 2x$$

Step 2: Rationalize the Denominator We still have a D in the denominator. Multiply numerator and denominator by the conjugate $(D + 3)$.

$$y_p = \frac{D+3}{(D-3)(D+3)} \cos 2x$$

$$y_p = \frac{D+3}{D^2 - 9} \cos 2x$$

Step 3: Substitute D^2 **again** Replace D^2 with -4 again.

$$y_p = \frac{D+3}{-4-9} \cos 2x$$

$$y_p = \frac{D+3}{-13} \cos 2x$$

Step 4: Operate Numerator Apply $(D + 3)$ to $\cos 2x$:

$$y_p = -\frac{1}{13}[D(\cos 2x) + 3\cos 2x]$$

* Derivative $D(\cos 2x) = -2\sin 2x$

$$y_p = -\frac{1}{13}[-2\sin 2x + 3\cos 2x]$$

Final Answer:

$$y_p = \frac{2\sin 2x - 3\cos 2x}{13}$$

Question 3: Case of Failure

Solve for y_p: $(D^2 + 4)y = \cos 2x$

Solution:

$$y_p = \frac{1}{D^2 + 4} \cos 2x$$

Step 1: Check Substitution Here $a = 2$, so $D^2 \to -4$. Denominator becomes $-4 + 4 = 0$. **Method Fails.**

Step 2: Apply Failure Rule Multiply by x and differentiate the denominator ($D^2 + 4 \to 2D$).

$$y_p = x \cdot \frac{1}{2D} \cos 2x$$

Step 3: Interpret $1/D$ The operator $\frac{1}{D}$ represents **Integration**.

$$y_p = \frac{x}{2} \int \cos 2x \, dx$$

Step 4: Integrate

$$\int \cos 2x \, dx = \frac{\sin 2x}{2}$$

$$y_p = \frac{x}{2} \left(\frac{\sin 2x}{2} \right)$$

Final Answer:

$$y_p = \frac{x \sin 2x}{4}$$

Next Method: Sometimes we have a product of two functions, like $e^x \sin x$. For this, we use the Shift Rule. Proceed to Method 4: Shift & Product Rules.